Rev	Date	Eng	Changes
AA	2026-10-07	NB	First release, derived from Sofar mote 000639-AB

nereus_Pi_shield_ADIN2111

Variant: Standard

P/N:PCB-000001-AA

Index

1 Cover
2 Top-Level
3 ADIN2111 (Single-Pair Ethernet)
4 Load Switch (payload port)
5 PoDL (Power over Data Line)
6 Bus Power Monitor
7 Power (buck converters)

Pi shield renders: TBD (after layout)

I2c Address Table

Device	I2C Address
Power Monitor	0x41

DIMENSIONS: TBD (after layout)

Top-Level Schematic
File: BM_Mote_1_Master.kicad_sch



The Bristlemouth Mote Board is licensed under a Creative Commons Attribution-NoDerivatives 4.0 International License. The Bristlemouth trademarks and the Bristlemouth logo are registered and unregistered marks belonging to Sofar Ocean Technologies, Inc., and are not included under the Creative Commons license.
https://creativecommons.org/licenses/by-nd/4.0/legalcode

Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /
File: nick_raw_routing.kicad_sch

Title: Cover Page

Size: A3	Date: 2026-10-07	**Rev: AA**
KiCad E.D.A. 10.0.6		Id: 1/7

I2C1 GPIO2/3 = LiFePO4wered only (0x43). Pi's 1.8k pull-ups.

I2C0 GPIO0/1 = shield: INA232 0x41, ADS1115 0x48 (bluewired). R26/R27 4.7k to PI_3V3.
Keep 0x50 free (HAT EEPROM probe at boot).

I2C3 GPIO22/27 = software I2C, Qwiic. Qwiic power and pull-ups on PI_3V3.

J2 breakout 3V3 pins (9, 11) = PI_3V3. Shield 3V3 is for shield parts only.

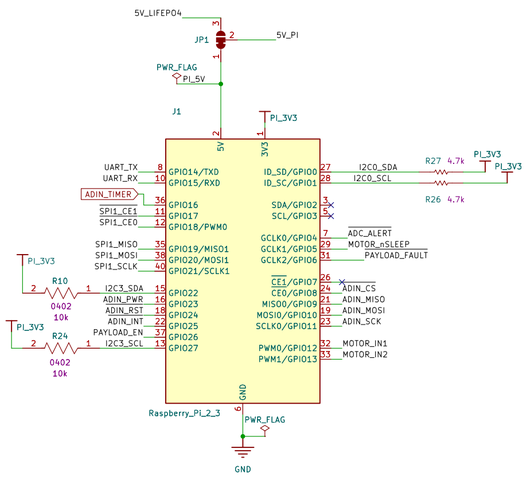
ADS1115: AIN0 = MP6550 VISEN (200 mV/A). AIN1 = shaft pot. AIN2 = spare. AIN3 unused.

JP1 power mode (middle pad = shield 5V. 5V_PI):

Mode 1 (as built): pads 1–2 bridged -> shield powers the PI through J1 pin 2.

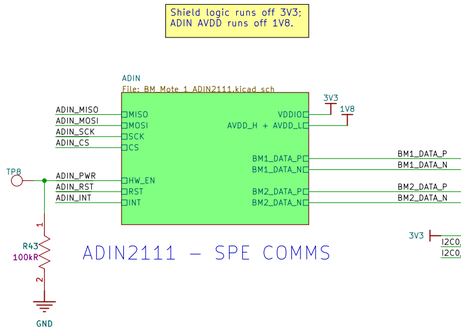
Mode 2 (LiFePO4wered): cut 1–2, bridge 2–3 -> shield 5V goes out on the JST-PH
to the LiFePO4wered INPUT. Never wire it to the LiFePO4wered 5V output.

Never plug a 5 V source into the PI's USB while 1–2 is bridged.

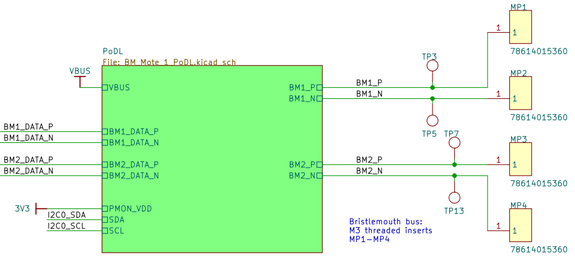


RASPBERRY PI — 40-PIN HEADER

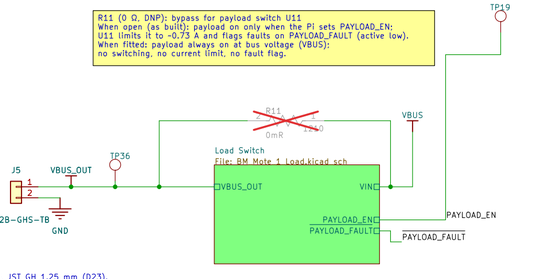
Shield logic runs off 3V3;
ADIN AVDD runs off 1V8.



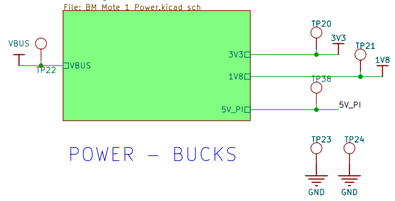
ADIN2111 — SPE COMMS



PoDL — POWER
OVER DATA LINE



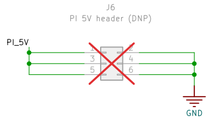
LOAD SWITCH — PAYLOAD POWER



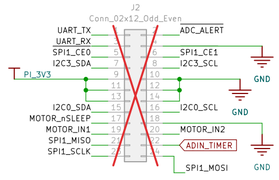
POWER — BUCKS

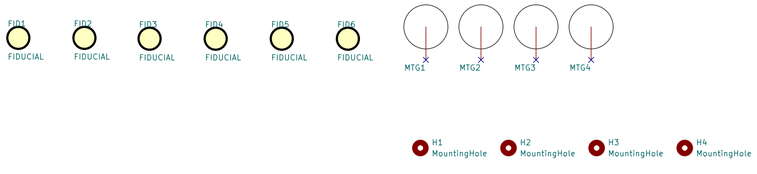


PI 3V3 REGULATED POWER

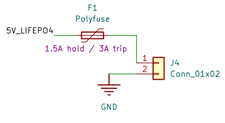


HEADER BREAKOUT FOR DEVELOPMENT

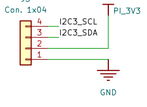




POWER PATH FOR EXTERNAL SHIELD



Qwiic CONNECTOR / I2C3 SOFTWARE EMULATION



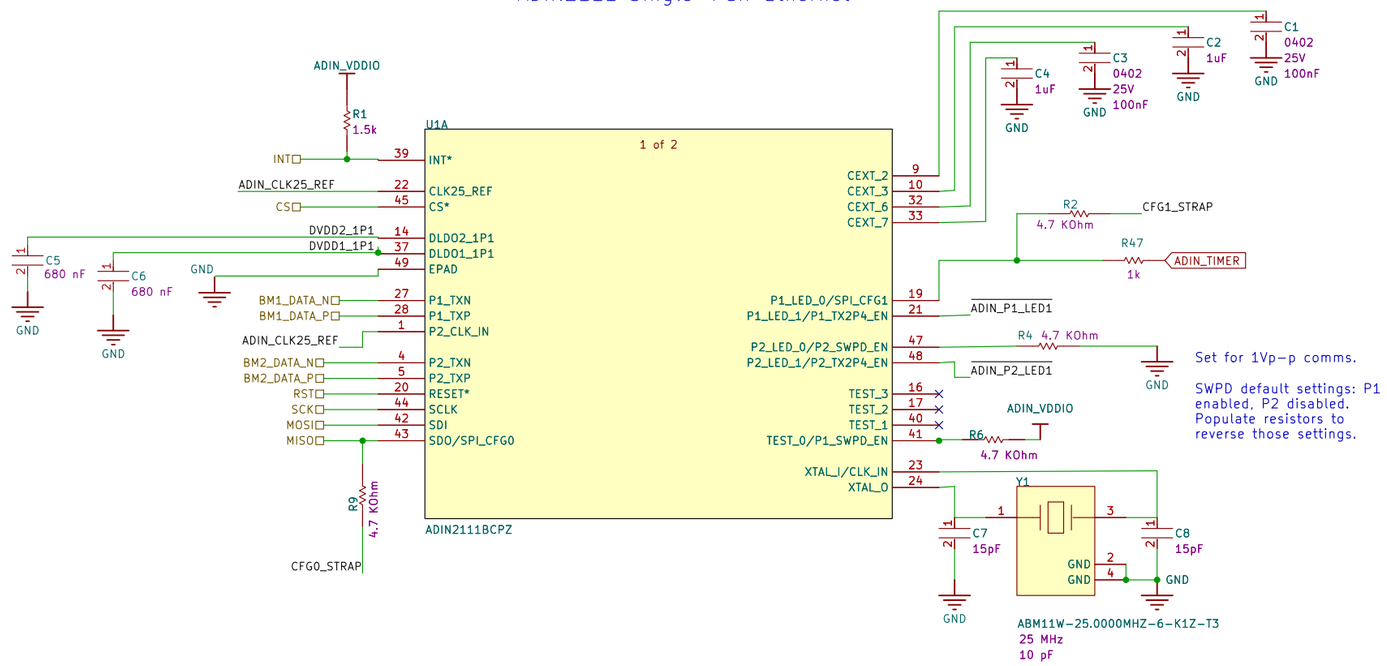
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/
File: BM_Mote_1_Master.kicad_sch

Title: Top-Level

Size: A2 | Date: 2026-10-07 | **Rev: AA**
KiCad E.D.A. 10.0.6 | Id: 2/7

ADIN2111 Single-Pair Ethernet



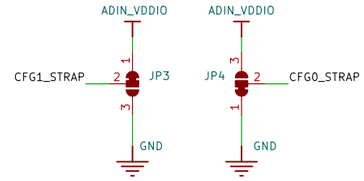
Set for 1Vp-p comms.

SWPD default settings: P1 enabled, P2 disabled. Populate resistors to reverse those settings.

ADIN2111 SPI CFG Straps

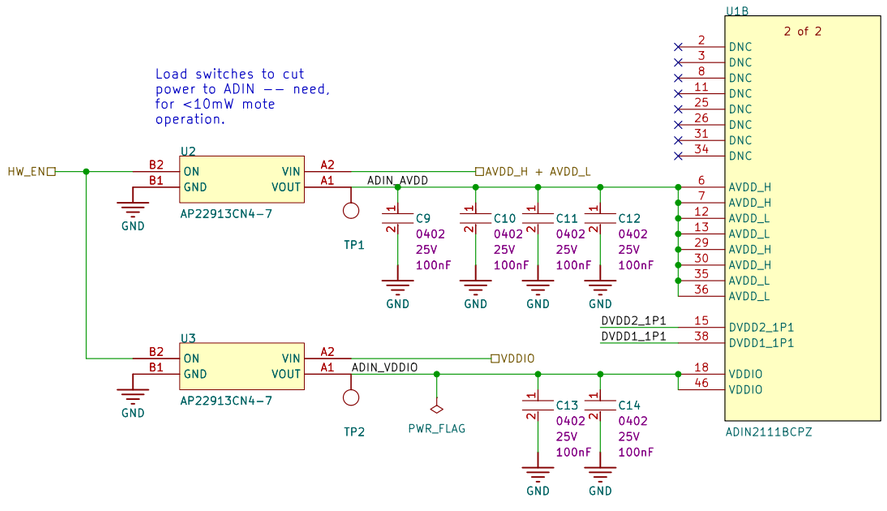
SPI CFG Mode	JP3 (SPI_CFG1, pin 19)	JP4 (SPI_CFG0, pin 43)
Generic SPI + CRC (default)	1-2 (VDDIO)	1-2 (GND)
Generic SPI, no CRC	1-2 (VDDIO)	2-3 (VDDIO)
OPEN Alliance + protection	2-3 (GND)	1-2 (GND)
OPEN Alliance, no protection	2-3 (GND)	2-3 (VDDIO)

SPI_CFG1 (pin19) defaults "1"
cut trace and solder bridge to get "0"

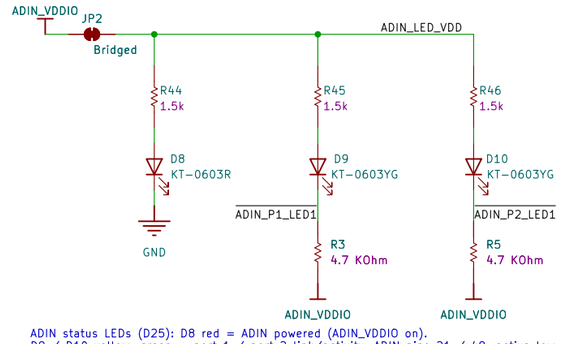


CFG0_STRAP (pin43) defaults "0".
cut trace and solder bridge to get "1"

ADIN2111 supplies: U2 (AVDD) and U3 (VDDIO) switched on by ADIN_PWR



ADIN2111 status LEDs: power, port 1 / port 2 link-activity



ADIN status LEDs (D25): D8 red = ADIN powered (ADIN_VDDIO on).
D9 / D10 yellow-green = port 1 / port 2 link/activity, ADIN pins 21 / 48, active low.
Off until the Pi enables each port's LED_1 over SPI; port 2 is also asleep until woken.
R3 / R5 keep the TX2P4_EN straps high (1.0 V p-p). Cut JP2 to disable all three LEDs.

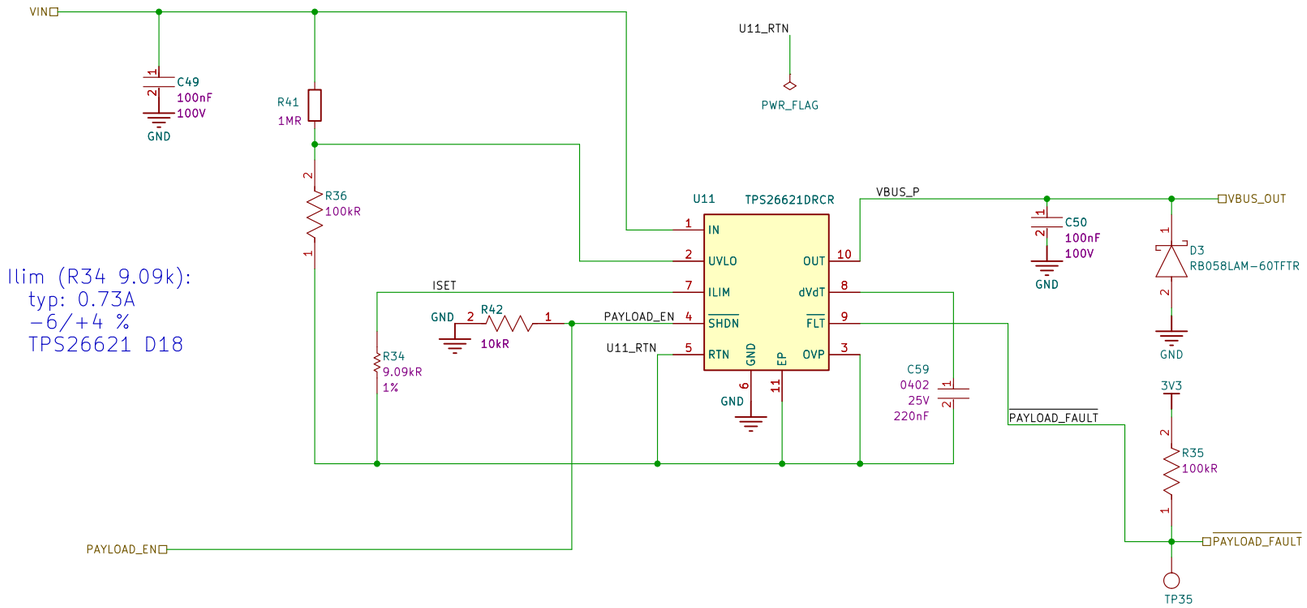
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/ADIN/
File: BM_Mote_1_ADIN2111.kicad_sch

Title: ADIN2111

Size: A3	Date: 2026-10-07	**Rev: AA**
KiCad E.D.A. 10.0.6		Id: 3/7

Load Switch for VBUS



Ilim (R34 9.09k):
typ: 0.73A
−6/+4 %
TPS26621 D18

Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

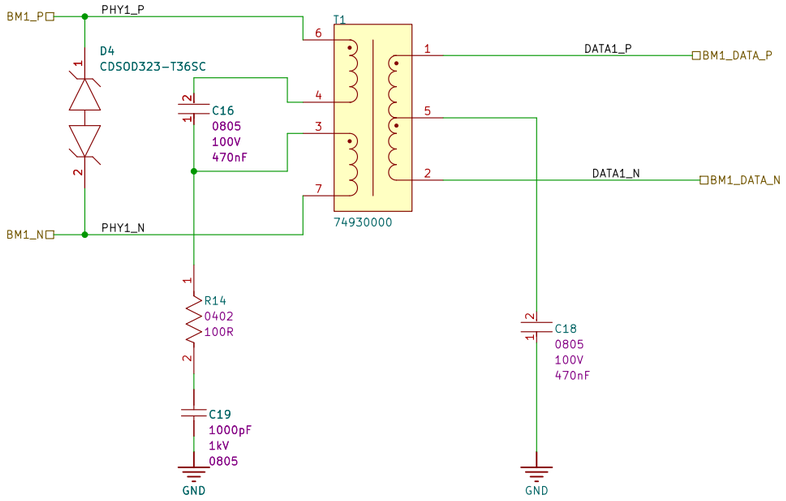
Sheet: /Top-Level Schematic/Load Switch/
File: BM_Mote_1_Load.kicad_sch

Title: Load Switch

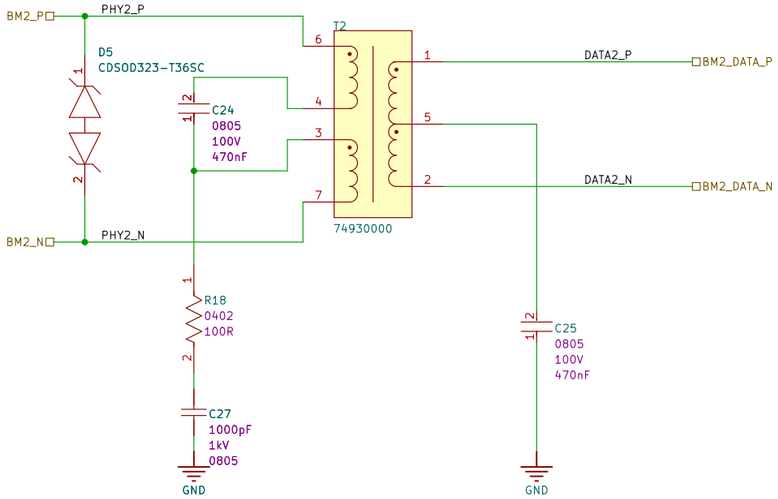
Size: A3	Date: 2026-10-07	**Rev: AA**
KiCad E.D.A. 10.0.6		Id: 4/7

Power Over Dataline

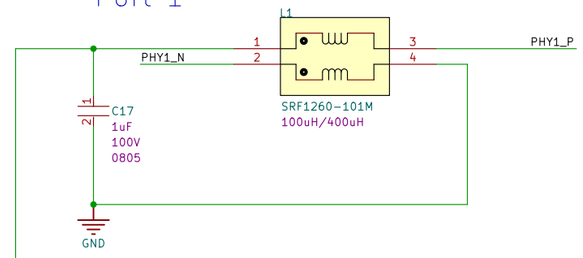
Coupling Network – Port 1



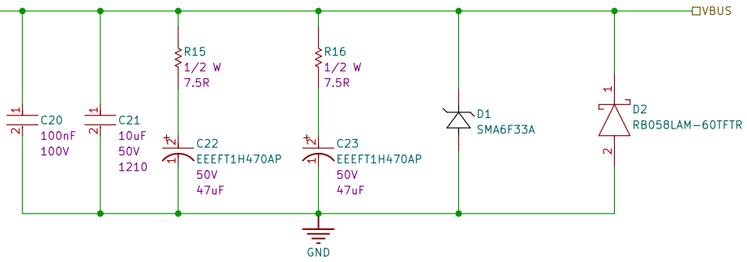
Coupling Network – Port 2



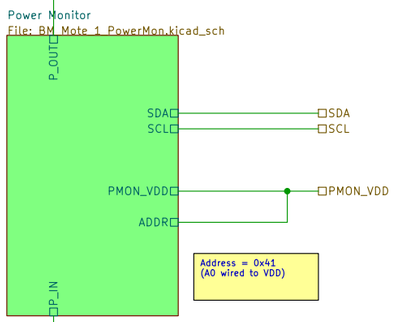
Port 1



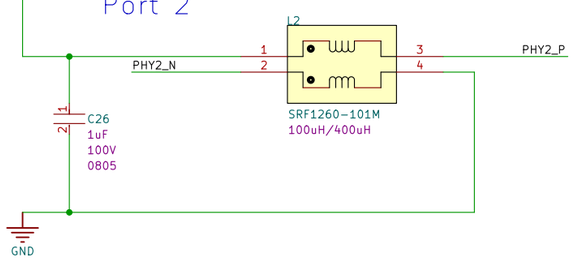
DC Connection + Circuit protections



Snubber resistors must be at least 1/2 W, to support hot-plugging.



Port 2



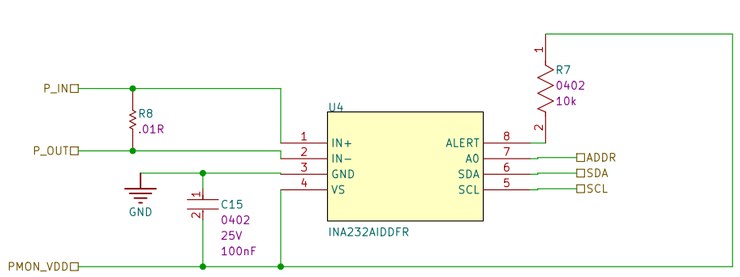
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/
File: BM_Mote_1_PoDL.kicad_sch

Title: PoDL (Power over Data Line)

Size: A3	Date: 2026-10-07	**Rev: AA**
KiCad E.D.A. 10.0.6		Id: 5/7

I2C0 (Pi GPIO0/1) · Address 0x41 (A0 = VS)



Power Monitor

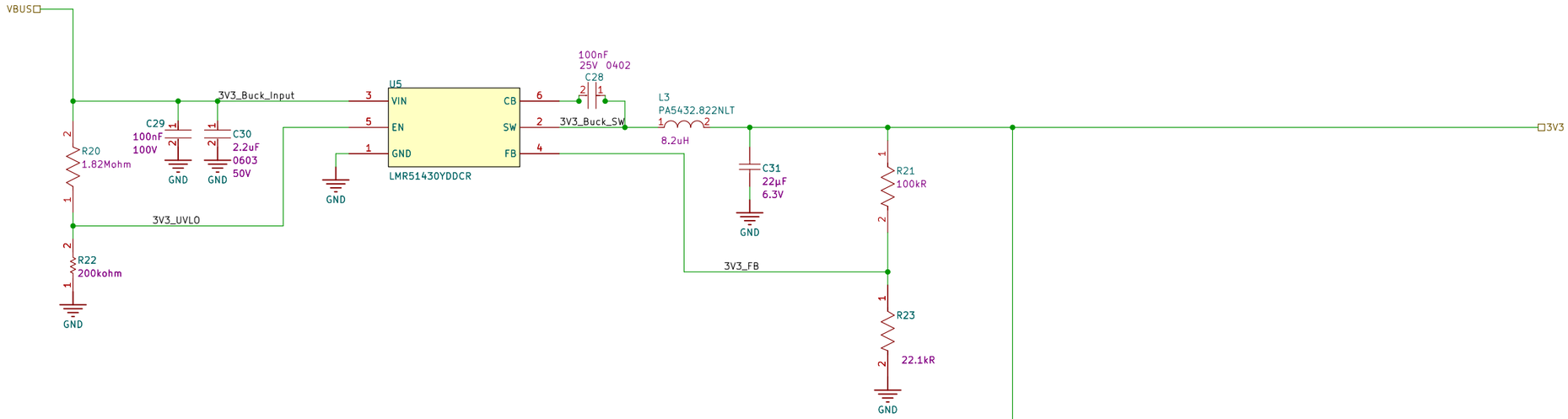
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/Power Monitor/
File: BM_Mote_1_PowerMon.kicad_sch

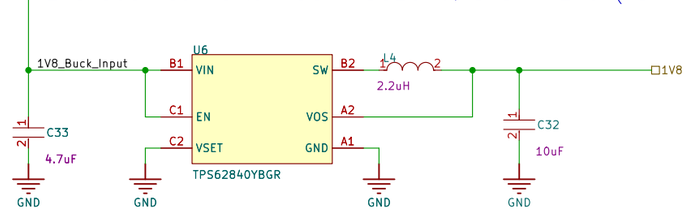
Title: Bus Power Monitor

Size: A3	Date: 2026-10-07	**Rev: AA**
KiCad E.D.A. 10.0.6		Id: 6/7

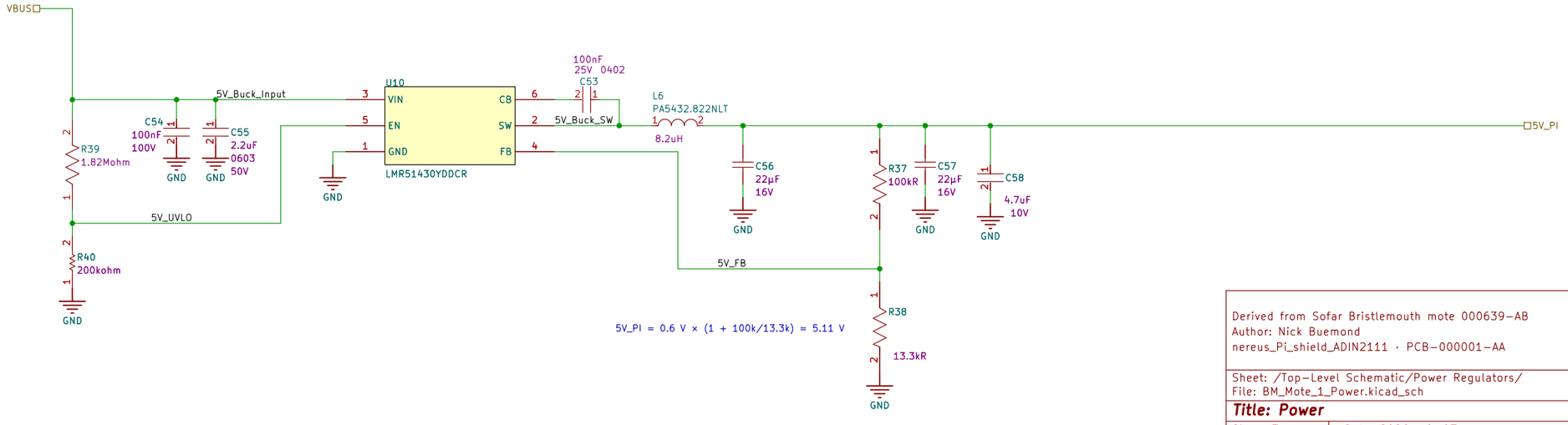
3V3 buck converter: U5 LMR51430, VBUS → 3V3



1V8 buck converter: U6 TPS62840, 3V3 → 1V8 (ADIN AVDD)



5V buck converter for the Pi: U10 LMR51430, VBUS → 5V_PI



Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
nereus_Pi_shield_ADIN2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/Power Regulators/
File: BM_Mote_1_Power.kicad_sch

Title: Power

Size: A3	Date: 2026-10-07	**Rev: AA**
KiCad E.D.A. 10.0.6		Id: 7/7